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054323 - INTERNET OF THINGS

POLITECNICO DI MILANO

COMPUTER SCIENCE AND ENGINEERING

IoT Challenge 4 - Homework
Exercise 1

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1 Exercise 1

1.1 System Overview

The system under analysis is an IEEE 802.15.4-based vibration monitoring network operating in **beacon-enabled mode, CFP only**. It consists of one PAN coordinator and three ESP32-based vibration monitoring nodes. Each node samples an accelerometer at 100 Hz, counts vibration peaks over a 10-second sliding window and transmits a payload whose size depends on the number of detected peaks. The network parameters are summarised in Table 1.

Parameter	**Value**
Nominal bit rate R	250 kbps
Packet size L	128 bytes
Reporting period T_{rep}	10 seconds
Vibration peak rate λ	0.4 peaks/window (Poisson)
Number of monitoring nodes	3
Operating mode	Beacon-enabled, CFP only

Table 1: System parameters

The payload size produced by each node per window is determined by the following logic (extracted from the pseudocode):

Condition	**Payload Size**	**Packets (n)**	**Output Rate**
0 peaks detected	512 bytes	4	r_0
1 peak detected	2 KB = 2048 bytes	16	r_1
>1 peaks detected	4 KB = 4096 bytes	32	r_2

Table 2: Output rate and packet count as a function of detected vibration peaks (n = payload/L, $L = 128$ bytes)

1.2 Question 1

1.2.1 Assumptions

- The number of vibration peaks per 10-second window follows a Poisson distribution with rate $\lambda = 0.4$ peaks/window.
- Each window is independent.
- The three output rates r_0, r_1, r_2 correspond to exactly 0, exactly 1 and more than 1 detected peaks, respectively.

1.2.2 Poisson PMF

For a Poisson random variable $K \sim \text{Poisson}(\lambda)$:

$$P(K = k) = \frac{e^{-\lambda}\lambda^k}{k!} \tag{1}$$

1.2.3 Output Rate Probabilities

Case r_0 - zero peaks detected:

$$P(r = r_0) = P(K = 0) = e^{-0.4} = \mathbf{0.6703} \tag{2}$$

Case r_1 - exactly one peak detected:

$$P(r = r_1) = P(K = 1) = e^{-0.4} \cdot 0.4 = \mathbf{0.2681} \tag{3}$$

Case r_2 - more than one peak detected:

$$P(r = r_2) = 1 - P(K = 0) - P(K = 1) = 1 - 0.6703 - 0.2681 = \mathbf{0.0616} \tag{4}$$

Results

$$P(r = r_0) = 0.6703, \quad P(r = r_1) = 0.2681, \quad P(r = r_2) = 0.0616$$

1.3 Beacon Interval Derivation

1.3.1 Standard Formula

In IEEE 802.15.4 beacon-enabled mode the Beacon Interval (BI) is not freely chosen to match the reporting period; it is derived from the **minimum output bit rate** of the node using the standard formula:

$$BI = \frac{L}{r_{\min}} \tag{5}$$

where L is the packet size in bits and $r_{\min}$ is the minimum required data rate in bit/s.

1.3.2 Minimum Output Rate

The minimum payload is generated in state r_0 (0 peaks detected): 512 bytes every 10 seconds.

$$r_{\min} = \frac{512 \text{ bytes} \times 8 \text{ bit/byte}}{10 \text{ s}} = \frac{4096}{10} = 409.6 \text{ bps} \tag{6}$$

1.3.3 Beacon Interval

$$BI = \frac{L \times 8}{r_{\min}} = \frac{128 \times 8}{409.6} = \frac{1024}{409.6} = \mathbf{2.5}\ \text{s} \tag{7}$$

1.3.4 Beacon Intervals per Reporting Window

The reporting period $T_{\text{rep}} = 10$ s spans multiple beacon intervals:

$$N_{BI} = \frac{T_{\text{rep}}}{BI} = \frac{10}{2.5} = \mathbf{4} \quad \text{beacon intervals per window} \tag{8}$$

This means each node's worst-case payload of 32 packets is **spread across 4 consecutive beacon intervals**, with a fixed fraction transmitted in each BI.

1.4 Question 2

1.4.1 Assumptions

- One packet of size $L = 128$ bytes fits exactly in one CFP slot.
- The CFP is dimensioned on the **worst-case** payload (r_2: 4 KB = 32 packets per window), ensuring that even in the most demanding scenario no slot starvation occurs.
- Since the worst-case payload is distributed over $N_{BI} = 4$ beacon intervals, only $32/4 = 8$ slots per node are required per beacon interval.
- All three nodes are statistically identical (same λ).

1.4.2 Worst-Case Slots per Node per Beacon Interval

$$n_{\text{worst/BI}} = \frac{n_2}{N_{BI}} = \frac{32}{4} = \mathbf{8}\ \text{slots/node/BI} \tag{9}$$

1.4.3 Total CFP Slot Assignment

Each node is assigned 8 guaranteed time slots (GTS) per beacon interval. With 3 nodes:

$$N_{\text{slots}}^{\text{CFP}} = 3 \times 8 = \mathbf{24}\ \text{slots} \tag{10}$$

Node	GTS slots / BI	Basis
Node 1	8	32 pkts (worst case)/4 BI = 8
Node 2	8	32 pkts (worst case)/4 BI = 8
Node 3	8	32 pkts (worst case)/4 BI = 8
Total CFP	**24**	

Table 3: Consistent CFP slot assignment per beacon interval (worst-case dimensioning)

1.5 Question 3

1.5.1 Slot Duration T_s

Each slot carries exactly one packet of $L = 128$ bytes at bit rate $R = 250$ kbps:

$$T_s = \frac{L \times 8}{R} = \frac{128 \times 8}{250000} = \frac{1024}{250000} = \mathbf{4.096}\ \text{ms} \tag{11}$$

1.5.2 Superframe Structure and N_{slots}

In IEEE 802.15.4 beacon-enabled mode, every superframe (= every BI) begins with a **beacon** transmitted by the PAN coordinator. The beacon occupies one slot duration T_s. It is followed by the CFP, which contains the 24 GTS slots. The total active slots per BI:

$$N_{\text{slots}} = \underbrace{1}_{\text{beacon}} + \underbrace{24}_{\text{CFP}} = \mathbf{25}\ \text{slots} \tag{12}$$

1.5.3 Active Period T_{active}

$$T_{\text{active}} = N_{\text{slots}} \times T_s = 25 \times 4.096\ \text{ms} = \mathbf{102.4}\ \text{ms} \tag{13}$$

1.5.4 Inactive Period T_{inactive}

The superframe period equals the beacon interval $BI = 2.5$ s:

$$T_{\text{inactive}} = BI - T_{\text{active}} = 2500 - 102.4 = \mathbf{2397.6}\ \text{ms} \tag{14}$$

1.5.5 System Duty Cycle

The duty cycle is computed over one beacon interval:

$$\text{DC} = \frac{T_{\text{active}}}{BI} = \frac{102.4}{2500} = 0.04096 \approx \mathbf{4.096}\% \quad \text{(as a fraction: } \mathbf{0.041}\text{)} \tag{15}$$

Results - Timing Parameters

BI	$= 2.5$ s
T_s	$= 4.096$ ms
$N_{\text{slots}}^{\text{CFP}}$	$= 24$ slots per BI
N_{slots}	$= 25 \quad (1 \text{ beacon} + 24 \text{ CFP})$
T_{active}	$= 102.4$ ms
T_{inactive}	$= 2397.6$ ms
Duty Cycle	$= 0.04096 \approx 0.041 \quad (4.096\%)$

1.6 Question 4

1.6.1 Constraint

We require the duty cycle to remain strictly below 10%. With N_{nodes} monitoring nodes, each assigned 8 CFP slots per BI, plus 1 beacon slot:

$$\text{DC} = \frac{(1 + N_{\text{nodes}} \times 8) \times T_s}{BI} < 10\% \tag{16}$$

1.6.2 Derivation

Substituting known values:

$$(1 + N_{\text{nodes}} \times 8) \times 4.096 \text{ ms} < 0.10 \times 2500 \text{ ms} = 250 \text{ ms} \tag{17}$$

$$1 + N_{\text{nodes}} \times 8 < \frac{250}{4.096} = 61.04 \tag{18}$$

$$N_{\text{nodes}} \times 8 < 60.04 \quad \Rightarrow \quad N_{\text{nodes}} < \frac{60.04}{8} = 7.505 \tag{19}$$

Therefore the maximum number of monitoring nodes is $N_{\text{nodes,max}} = 7$.

1.6.3 Additional Nodes

Since the system already contains 3 monitoring nodes:

$$\Delta N = N_{\text{nodes,max}} - N_{\text{current}} = 7 - 3 = \mathbf{4} \text{ additional nodes} \tag{20}$$

Result

Up to **4 additional vibration monitoring nodes** can be added while keeping the duty cycle strictly below 10%. The total node count would then be 7, yielding:

$$\mathrm{DC}_{\max} = \frac{(1 + 7 \times 8) \times 4.096}{2500} = \frac{57 \times 4.096}{2500} = \frac{233.472}{2500} = 9.339\% < 10\%$$

1.7 Summary of Results

Quantity	Value	Notes
$P(r = r_0)$	0.6703	0 peaks, 512 B payload
$P(r = r_1)$	0.2681	1 peak, 2 KB payload
$P(r = r_2)$	0.0616	>1 peaks, 4 KB payload
$r_{\min}$	409.6 bps	Minimum output rate (state r_0)
BI	2.5 s	Beacon interval ($= L{\times}8/r_{\min}$)
N_{BI}	4	BI per reporting window
GTS slots per node per BI	8	Worst-case 32 pkts / 4 BI
$N_{\text{slots}}^{\text{CFP}}$	24	3 nodes × 8 slots
N_{slots} (total/BI)	25	1 beacon + 24 CFP
T_s	4.096 ms	Slot duration
T_{active}	102.4 ms	Beacon + CFP per BI
T_{inactive}	2397.6 ms	Sleep period per BI
Duty Cycle	0.041 (4.096%)	Below 10%
Additional nodes allowed	4	DC constraint < 10%

Table 4: Summary of all computed quantities for Exercise 1